INTRODUCTION

In today's globalized economy, foreign exchange rates play a crucial role in influencing international trade, investment decisions, and overall economic stability. Fluctuations in currency values directly impact import-export businesses, financial markets, and policy planning, making it essential to analyse and understand exchange rate behaviour over time.

This project focuses on analysing the exchange rate trends of major global currencies—USD, EUR, GBP, JPY, and CNY—against the Indian Rupee (INR). The dataset has been structured and enriched with multiple time-based features such as year, month, and quarter, along with derived metrics like day-over-day percentage change to capture short-term movements.

Using this processed dataset, interactive dashboards were developed in Power BI to visualize trends, monitor fluctuations, and compare currency performance across different time periods. These dashboards provide a clear and intuitive representation of complex data, enabling users to gain meaningful insights into currency behaviour and market dynamics.

OBJECTIVE

- To analyse the exchange rate trends of major global currencies (USD, EUR, GBP, JPY, and CNY) against the Indian Rupee (INR) over a selected time period.
- To perform comparative analysis across multiple currencies in order to evaluate their relative performance.
- To utilize time-based features such as year, month, and quarter for effective time-series analysis.
- To develop interactive Power BI dashboards that visually represent currency movements and patterns.
- To enable dynamic data exploration through filters and slicers for better user interaction.
- To identify overall patterns and trends in exchange rate behaviour over time.
- To build and implement a machine learning model for predicting future exchange rates based on historical data.
- To achieve a highly accurate predictive model, with performance exceeding 95% accuracy, ensuring reliable forecasting outcomes based on evaluation metrics.

DATASET DESCRIPTION

The dataset used in this project consists of historical exchange rate data of major global currencies—USD, EUR, GBP, JPY, and CNY—against the Indian Rupee (INR). The data has been structured in a time-series format to support both analytical visualization and predictive modelling.

For the purpose of dashboard development, the dataset was organized into a structured format with key attributes including Date, Currency, and INR per unit value, along with derived time-based features such as Year, Month, Quarter, Month Name, and Day of Week. These features enable detailed time-series analysis and support interactive filtering within the Power BI dashboards.

In addition to the primary dataset, a calculated metric—day-over-day percentage change—was incorporated to capture short-term fluctuations and provide insights into currency volatility. This enhanced dataset serves as the foundation for visual analytics, allowing users to explore trends, compare currency performance, and identify patterns over time.

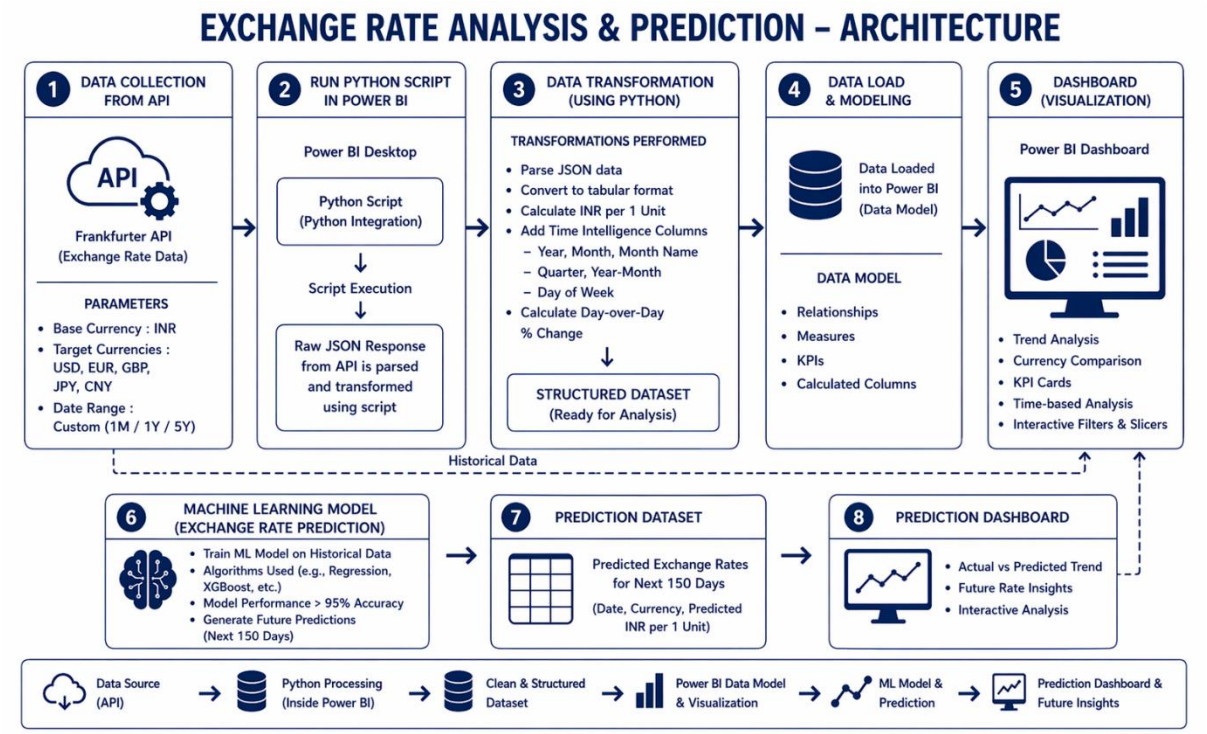
For the predictive analysis component, the dataset was further extended to include forecasted values for the next 150 days, generated using a machine learning model trained on historical exchange rate data. This extended dataset enables forward-looking analysis and supports the evaluation of future currency trends.

Overall, the project utilizes two integrated datasets:

- A historical dataset for exploratory analysis and dashboard visualization
- A prediction dataset for forecasting and advanced analytical insights

Together, these datasets provide a comprehensive framework for both descriptive and predictive analysis of currency exchange rates.

ARCHTECTURE



Methodology

The methodology of this project follows a structured pipeline, transforming raw exchange rate data into meaningful visual insights and predictive outcomes. The process integrates data collection, transformation, visualization, and machine learning to build a comprehensive analytical solution.

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1. Data Collection

The initial stage involves gathering historical exchange rate data for major global currencies (USD, EUR, GBP, JPY, and CNY) against the Indian Rupee (INR). The data is collected in a time-series format over a defined period, ensuring consistency and completeness for analysis.

The collected data serves as the foundational input for further processing and analysis.

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2. Data Extraction and Processing using Python in Power BI

To streamline the workflow, Python scripting is integrated directly within Power BI. The raw data is extracted and processed using Python, enabling efficient handling of structured and semi-structured data formats.

The script performs the following key operations:

- Parses raw data into a structured tabular format
- Converts exchange rates into INR per unit values
- Prepares the dataset for further transformation

Code Implementation:

```
import requests
import pandas as pd
from datetime import date, timedelta
from dateutil.relativedelta import relativedelta

# ── Config — change PERIOD to "1month", "1year", or "5year" ──
PERIOD = "1year"
TARGET_CURRENCIES = ["USD", "EUR", "GBP", "JPY", "CNY"]

# ── Date range ──────────────────────────────────────────────
yesterday = date.today() - timedelta(days=1)
if PERIOD == "1month":
    start_date = yesterday - relativedelta(months=1)
elif PERIOD == "1year":
    start_date = yesterday - relativedelta(years=1)
elif PERIOD == "5year":
    start_date = yesterday - relativedelta(years=5)
end_date = yesterday
```

```
# ── API Call ──────────────────────────────────────────────────────
url = f"https://api.frankfurter.dev/v1/{start_date}..{end_date}"
params = {
    "base":    "INR",
    "symbols": ",".join(TARGET_CURRENCIES),
}

response = requests.get(url, params=params)
data = response.json()

# ── Parse into long format ──────────────────────────────
rows = []
for date_str, rates in data.get("rates", {}).items():
    for currency in TARGET_CURRENCIES:
        rate = rates.get(currency)
        rows.append({
            "Date":          date_str,
            "Currency":      currency,
            "INR_per_1_Unit": round(1 / rate, 4) if rate else None,
        })

# ── Build DataFrame ──────────────────────────────────────────────
dataset = pd.DataFrame(rows)
dataset["Date"]      = pd.to_datetime(dataset["Date"])
dataset["Year"]      = dataset["Date"].dt.year
dataset["Month"]     = dataset["Date"].dt.month
dataset["MonthName"] = dataset["Date"].dt.strftime("%B")
dataset["Quarter"]   = dataset["Date"].dt.quarter.apply(lambda q: f"Q{q}")
dataset["YearMonth"] = dataset["Date"].dt.strftime("%Y-%m")
dataset["DayOfWeek"] = dataset["Date"].dt.day_name()

# Day-over-day % change per currency
dataset = dataset.sort_values(["Currency", "Date"]).reset_index(drop=True)
dataset["DayChange_pct"] = (
    dataset.groupby("Currency")["INR_per_1_Unit"]
    .pct_change() * 100
).round(4)

# ── Power BI reads the variable named 'dataset' ──
dataset
```

3. Data Transformation and Feature Engineering

The processed dataset is further enhanced by applying multiple transformation techniques to enable in-depth analysis. Time-based features are derived from the date field to support time-series exploration.

Key transformations include:

- Extraction of Year, Month, Quarter, and Day of Week
- Creation of Month Name and Year-Month for better visualization
- Calculation of Day-over-Day Percentage Change to measure currency volatility

These transformations improve the analytical capability of the dataset and make it suitable for visualization and modelling.

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4. Data Loading and Modelling in Power BI

The transformed dataset is loaded into Power BI, where a data model is created to support efficient querying and visualization.

This stage includes:

- Defining relationships (if applicable)
- Creating calculated columns and measures
- Designing key performance indicators (KPIs)

The data model ensures optimized performance and scalability of the dashboards.

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5. Dashboard Development and Visualization

Interactive dashboards are developed in Power BI to present insights in a clear and intuitive manner. Various visualizations are used to represent trends, comparisons, and patterns in the data.

Key features of the dashboard include:

- Trend analysis using line charts
- Comparative analysis across currencies
- Time-based filtering using slicers
- Interactive exploration of exchange rate movements

These dashboards enable users to gain a comprehensive understanding of currency behaviour

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6. Machine Learning Model for Prediction

To extend the analysis beyond historical data, a machine learning model is implemented to predict future exchange rates. The model is trained using historical time-series data and designed to capture underlying patterns and trends.

Key aspects of this stage include:

- Training the model on historical exchange rate data
- Evaluating model performance using appropriate metrics
- Achieving high predictive performance (above 95%)
- Generating future exchange rate predictions for the next 150 days

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7. Prediction Dataset and Visualization

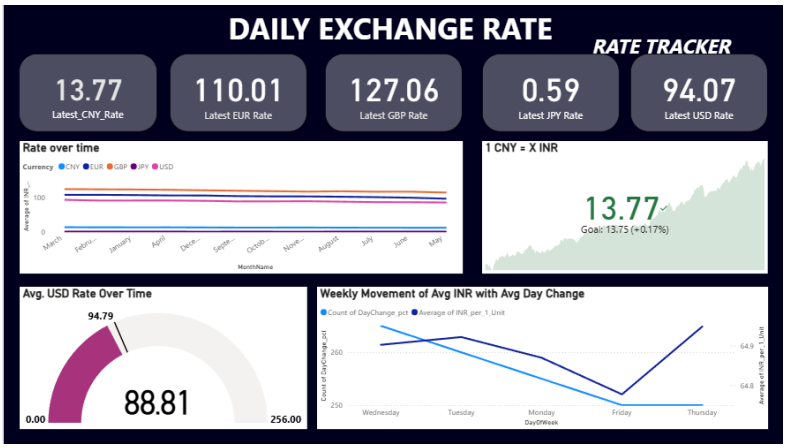
The predicted values are structured into a separate dataset, which is then integrated into Power BI for visualization.

This enables:

- Comparison between actual and predicted values
- Visualization of future trends
- Enhanced analytical insights through forecasting

VISUAL IMPLEMENTATION & INSIGHTS

DASHBOARD 1 – Rate Tracker & Real-Time Valuation



Visual	Fields Configuration
Line Chart – Rate over time	X axis – MonthName, Y axis – Average of INR_per_1_Unit, Legend – Currency
KPI – 1CNY = X INR	Value – Latest_CNY_Rate, Trend axis – Date , Target – Previous_CNY_Rate
Gauge – Avg USD Rate Over Time	Value – AVG PRICE, Maximum Value – Count of INR_per_1_Unit, Target Value – MAX Price
Line Chart – Weekly Movement Of Avg INR with Avg Day Change	X axis –DayOfWeek , Yaxis – Count of DayChange_pct , Secondary Y axis – AVG of INR_per_1_Unit
Card 1 – Latest CNY Rate	Fields – Latest CNY Rate
Card 2 – Latest EUR Rate	Fields – Latest EUR Rate
Card 3 – Latest GBP Rate	Fields– Latest GBP Rate
Card 4 – Latest JPY Rate	Fields – Latest JPY Rate
Card 5 – Latest USD Rate	Fields – Latest USD Rate

Focus: Current market value & historical context

- **Line Chart (Rate over time):** Reveals long-term appreciation/depreciation trends of each currency against INR.
  - ***Decision:*** If USD/INR is consistently rising, delay USD payables or execute hedge contracts early.
- **KPI (1 CNY = X INR):** Provides real-time valuation of Chinese Yuan.
  - ***Decision:*** Negotiate better pricing with Chinese suppliers when CNY is weak against INR.
- **Gauge (Avg USD Rate):** Compares current USD rate to historical max and average.
  - ***Decision:*** Increase hedge coverage (e.g., sell USD forwards) if current rate is near historical high.
- **Line Chart (Weekly Movement):** Identifies which weekday shows lowest volatility or average rate.
  - ***Decision:*** Schedule large forex transactions on historically stable days (e.g., Tuesday instead of Friday).
- **Latest Rate Cards:** Instant snapshot of all 5 currencies.

- ***Decision:*** Quickly prioritize hedging for the currency that has strengthened most in the last 24 hours.

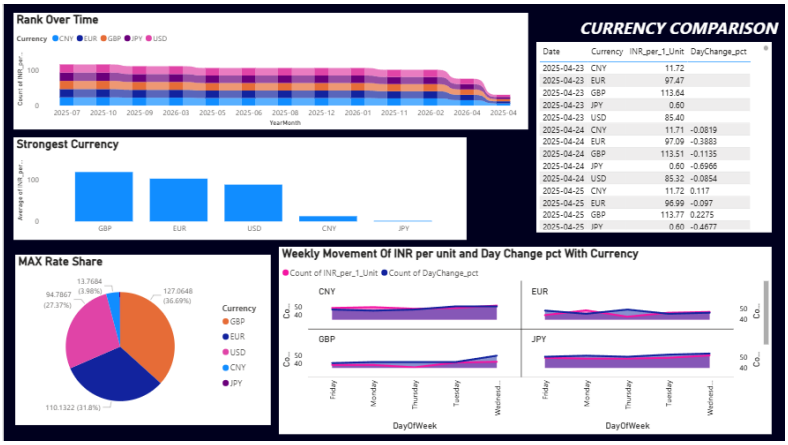
DASHBOARD 2 – Volatility & Short-Term Fluctuations

Visual	Fields Configuration
Line Chart – Daily % Change	X axis – MonthName, Y axis – Count of DayChange_pct, Legend – Currency
Matrix – Weekly overview heatmap	Rows – DayOfWeek, Columns – Currency, Values – Last DayChange_pct
Line and Clustered Column Chart – Average Monthly Rate Compared with EUR	X axis – MonthName, Column Y axis – Average of INR_per_1_Unit, Line y axis – Latest EUR Rate, Column Legend - Currency
Line and Clustered Column Chart – Weekly Movement Of INR Rates	X axis – DayOfWeek, Column Y axis – Count of INR_per_1_Unit
Line and Clustered Column Chart - Volatility vs Rate	X axis – Currency, Column Y axis – Count of DayChange_pct, Line y axis – Average of INR_per_1_Unit

Focus: Risk assessment & day-to-day market behaviour

- **Line Chart (Daily % Change):** Identifies which currency experiences highest daily volatility (e.g., GBP > CNY).
  - ***Decision:*** Avoid holding open positions in high-volatility currencies over weekends or before major economic data releases.
- **Matrix (Weekly Heatmap):** Shows which currency-day combination has extreme positive/negative movements.
  - ***Decision:*** Set dynamic stop-losses – tighter for Monday EUR, wider for Friday GBP.
- **Clustered Column Chart (Monthly Rate vs EUR):** Compares EUR monthly average against other currencies.
  - ***Decision:*** Identify short-term arbitrage opportunities between EUR and GBP.
- **Volatility vs Rate Chart:** Maps currencies as high/low rate combined with high/low volatility.
  - ***Decision:*** Allocate treasury funds toward low-volatility currencies for safety; use high-volatility currencies for active trading.

DASHBOARD 3 - Comparative Currency Strength & Ranking

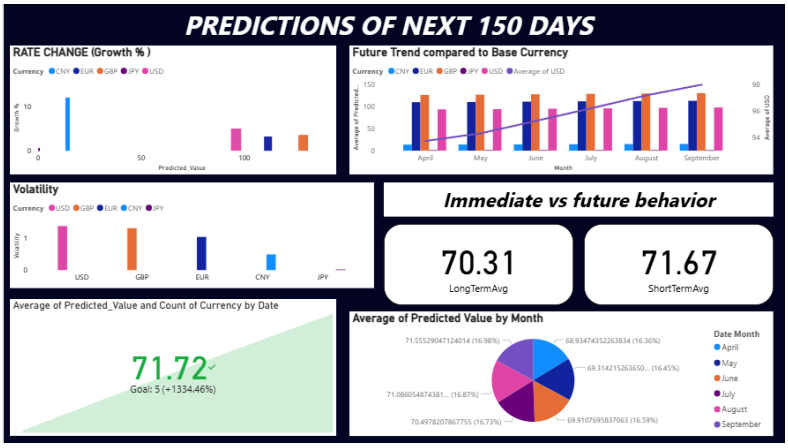


Visual	Fields Configuration
Ribbon Chart – Rate over time	X axis – YearMonth, Y axis – Count of INR_per_1_Unit, Legend – Currency
Stacked Column Chart – Strongest Currency	X axis – Currency , Y axis – Average of INR_per_1_Unit
Table	Columns – Date , Currency, INR_per_1_Unit , DayChange_pct
Pie Chart - MAX Rate Share	Legend – Currency , Values – Max of INR_Per_1_Unit
Area Chart – Weekly Movement Of INR per unit and Day Change pct With Currency	X axis – DayOfWeek, Y axis – Count of INR_per_1_Unit, Secondary Y axis – Count of DayChange_pct, Small multiples - Currency

Focus: Which currency is strongest/weakest against INR over time

- **Ribbon Chart (Rank over time):** Shows how currency rankings change monthly (e.g., GBP #1, JPY last).
  - ***Decision:*** Rebalance multi-currency portfolios monthly based on rank changes.
- **Stacked Column Chart (Strongest Currency):** Identifies the single strongest currency overall (typically GBP or USD).
  - ***Decision:*** Invoice exports in the strongest currency to maximize INR realization.
- **Pie Chart (Max Rate Share):** Highlights which currency reaches the highest peak values.
  - ***Decision:*** Use that currency as a benchmark for setting target INR rates in contracts.
- **Area Chart (Weekly Movement by Currency):** Shows day-wise rate + change intensity per currency.
- ***Decision:*** Route international payments through the currency showing the most stable weekly pattern.

DASHBOARD 4 – Predictive & Future Trend Analysis (ML Model)



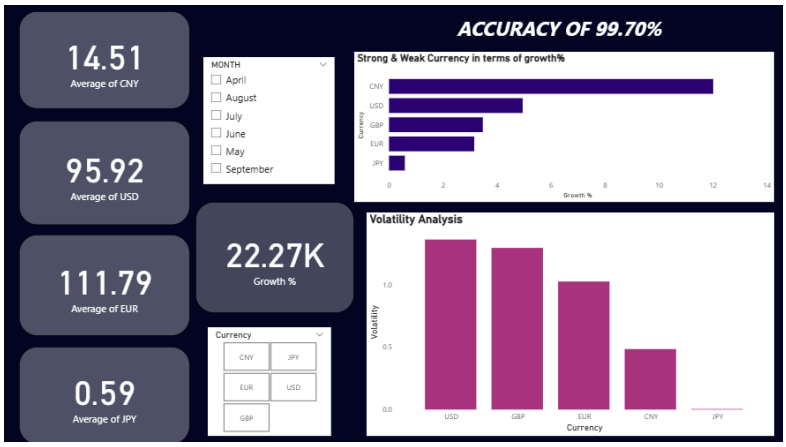
Visual	Fields Configuration
Clustered Column Chart – Rate Change (Growth %)	X axis – Predicted_value, Y axis – Growth % , Legend – Currency
Line and Clustered Column Chart – Future Trend Compared to Base Currency	X axis – Date Month, Column Y axis – Average of Predicted_Value, Line Y axis – Average of USD, Column Legend – Currency
Clustered Column Chart - Volatility	X axis – Currency, Y axis –Volatility, Legend – Currency
Card 1 – Long Term Avg	Fields – Long Term Avg
Card 2 – Short Term Avg	Fields – Short Term Avg
KPI – Average of Predicted_Value and Count of Currency	Value – Average of Predicted_Value, Trend axis – Date, Target – Count of Currency
Pie Chart – Average of Predicted Value by Month	Value - Average of Predicted_Value, Legend – Date Month

Focus: Forecasting future market value (150-day horizon with 99.7% accuracy)

- **Clustered Column Chart (Growth %):** Shows predicted % appreciation/depreciation per currency.
  - ***Decision:*** Pre-buy foreign currency if prediction shows sharp INR depreciation for a particular currency.
- **Line Chart (Future Trend vs Base):** Compares predicted future rate against a base currency (e.g., USD).
  - ***Decision:*** Structure loan repayments – if EUR is predicted to rise faster than USD, repay EUR loans early.
- **Volatility Chart (Predicted):** Forecasts which currency will become more volatile.
  - ***Decision:*** Increase hedge ratio for high-predicted-volatility currencies from 50% → 70%.
- **KPI (Short-term Avg vs Long-term Avg):** Compares 7-day average against overall predicted average.
  - ***Decision:*** Execute trades immediately if short-term avg is significantly lower than long-term avg (trading at a discount).

- **Pie Chart (Predicted Value by Month):** Shows which month will have highest predicted rates.
- ***Decision:*** Time major forex conversions (e.g., annual royalty payments) to the month with lowest predicted rate.

DASHBOARD 5 - Filtered Deep Dive & Sensitivity Analysis



Visual	Fields Configuration
Card 1 – Average of CNY	Fields – Average of CNY
Card 2 – Average of USD	Fields – Average of USD
Card 3 – Average of EUR	Fields – Average of EUR
Card 4 – Average of JPY	Fields – Average of JPY
Card 5 – Growth %	Fields – Growth %
Cluster Bar Chart – Strong & Weak Currency in terms of growth %	Y axis – Currency, X axis – Growth %
Clustered Column Chart - Volatility Analysis	X axis – Currency, Y axis - Volatility
Slicer	Month
Slicer	Currency

Focus: Customizable scenario analysis using slicers

- **Cluster Bar Chart (Strong & Weak by Growth %):** Shows which currency has highest positive vs negative predicted growth.
  - ***Decision:*** Go long on high-growth currencies, short on negative-growth currencies in forex derivatives.
- **Volatility Analysis Chart:** Compares volatility across currencies with applied filters.
  - ***Decision:*** Adjust margin requirements for international subsidiaries based on filtered currency volatility.
- **Slicers (Month + Currency):** Allows isolation of specific time periods and currencies.
  - ***Decision:*** Run "what-if" scenarios – e.g., "What was JPY behaviour only in Q2 of last year?" to inform seasonal strategies.
- **Average Cards (CNY, USD, EUR, JPY):** Quick reference for filtered averages.

- ***Decision:*** Compare filtered averages to unfiltered benchmarks to identify market anomalies or mispricings.

SUMMARY

- **USD/INR at 3-month high & rising** → Delay USD payments, buy only essential forwards.
- **GBP has highest daily volatility** → Use options (not forwards) to hedge GBP exposure.
- **JPY consistently ranked weakest** → Invoice Japanese exports in INR instead of JPY.
- **ML model predicts EUR depreciation >2%** → Shift EUR receivables to USD or GBP.
- **Wednesday shows lowest volatility** → Schedule all large corporate forex settlements on Wednesdays.
- **CNY stable but EUR volatile** → Fund Chinese imports via EUR → CNY conversion for arbitrage.
  Short-term avg < long-term avg (discount) → Execute trades immediately before prices rise.

DAX MEASURES

```
AVG DAY CHANGE USD = AVERAGE('dataset'[DayChange_pct] )
```

```
AVG DAYCHANGEpct = AVERAGE('dataset'[DayChange_pct])
```

```
AVG PRICE = AVERAGE('dataset'[INR_per_1_Unit])
```

```
CNY_DayChange_Latest =
    VAR LatestDate =
    CALCULATE(
    MAX('dataset'[Date]),
    'dataset'[Currency] = "CNY"
    )
    RETURN
    CALCULATE(
    MAX('dataset'[DayChange_pct]),
    'dataset'[Currency] = "CNY",
    'dataset'[Date] = LatestDate
    )
```

```
INR_per_unit = DISTINCT('dataset'[INR_per_1_Unit])
```

```
Latest EUR Rate =
VAR LatestDate =
CALCULATE(
MAX('dataset'[Date]),
'dataset'[Currency]= "EUR"
)
RETURN
CALCULATE(
MAX('dataset'[INR_per_1_Unit]),
'dataset'[Currency] = "EUR",
'dataset'[Date] = LatestDate
)
```

```
Latest GBP Rate =
VAR LatestDate =
CALCULATE(
MAX('dataset'[Date]),
'dataset'[Currency]= "GBP"
)
RETURN
CALCULATE(
MAX('dataset'[INR_per_1_Unit]),
'dataset'[Currency] = "GBP",
'dataset'[Date] = LatestDate
)
```

```
Latest JPY Rate =
VAR LatestDate =
CALCULATE(
MAX('dataset'[Date]),
'dataset'[Currency]= "JPY"
)
```

```
RETURN
CALCULATE(
MAX('dataset'[INR_per_1_Unit]),
'dataset'[Currency] = "JPY",
'dataset'[Date] = LatestDate
)
```

Latest USD Rate =

```
VAR LatestDate =
CALCULATE(
MAX('dataset'[Date]),
'dataset'[Currency]= "USD"
)
RETURN
CALCULATE(
MAX('dataset'[INR_per_1_Unit]),
'dataset'[Currency] = "USD",
'dataset'[Date] = LatestDate
)
```

Latest_CNY_Date =

```
CALCULATE(
MAX('dataset'[Date]),
'dataset'[Currency] = "CNY"
)
```

Latest_CNY_Rate =

```
VAR LatestDate =
CALCULATE(
MAX('dataset'[Date]),
'dataset'[Currency]= "CNY"
)
RETURN
CALCULATE(
MAX('dataset'[INR_per_1_Unit]),
'dataset'[Currency] = "CNY",
'dataset'[Date] = LatestDate
)
```

MAX Price = MAX('dataset'[INR_per_1_Unit])

Previous_CNY_Date =

```
VAR LatestDate =
CALCULATE(
MAX('dataset'[Date]),
'dataset'[Currency]= "CNY"
)
RETURN
CALCULATE(
MAX('dataset'[Date]),
'dataset'[Currency]= "CNY",
'dataset'[Date] < LatestDate
)
```

Previous_CNY_Rate =

```
VAR LatestDate =
CALCULATE(
```

```
    MAX('dataset'[Date]),
    'dataset'[Currency] = "CNY"
    )

    VAR PrevDate =
    CALCULATE(
    MAX('dataset'[Date]),
    'dataset'[Currency] = "CNY",
    'dataset'[Date] < LatestDate
    )

    RETURN
    CALCULATE(
    MAX('dataset'[INR_per_1_Unit]),
    'dataset'[Currency]= "CNY",
    'dataset'[Date]= PrevDate
    )
```

```
    USD_DayChange =
    CALCULATE(
    AVERAGE('dataset'[DayChange_pct]),
    'dataset'[Currency] = "USD"
    )
```

```
    USD_DayChange_Latest =
    VAR LatestDate =
    CALCULATE(
    MAX('dataset'[Date]),
    'dataset'[Currency] = "USD"
    )
    RETURN
    CALCULATE(
    MAX('dataset'[DayChange_pct]),
    'dataset'[Currency] = "USD",
    'dataset'[Date] = LatestDate
    )
```

```
        Growth % =
VAR FirstValue =
  CALCULATE(
    FIRSTNONBLANK('final_forecast'[Predicted_Value], 1),
    ALLEXCEPT('final_forecast', 'final_forecast'[Currency])
  )

VAR LastValue =
  CALCULATE(
    LASTNONBLANK('final_forecast'[Predicted_Value], 1),
    ALLEXCEPT('final_forecast', 'final_forecast'[Currency])
  )

RETURN
  DIVIDE(LastValue - FirstValue, FirstValue) * 100
```

LongTermAvg = AVERAGE('final_forecast'[Predicted_Value])
Growth %% = DIVIDE([Latest Value] - [Start Value], [Start Value]) * 100
Latest Value = CALCULATE(LASTNONBLANK('final_forecast'[Predicted_Value], 1), FILTER('final_forecast', 'final_forecast'[Date] = MAX('final_forecast'[Date])))
ShortTermAvg = CALCULATE(AVERAGE('final_forecast'[Predicted_Value]), DATESINPERIOD('final_forecast'[Date], MAX('final_forecast'[Date]), -7, DAY))
Start Value = CALCULATE(FIRSTNONBLANK('final_forecast'[Predicted_Value], 1), FILTER('final_forecast', 'final_forecast'[Date] = MIN('final_forecast'[Date])))
Volatility = STDEV.P('final_forecast'[Predicted_Value])
MONTH = FORMAT('wide_forecast'[Date], "MMMM") → **New Column created.**

Machine Learning Model for Exchange Rate Prediction

To extend the scope of analysis beyond historical trends, a machine learning–based predictive model was developed to forecast future exchange rates of selected currencies against the Indian Rupee (INR). The objective of this model is to capture underlying patterns in historical time-series data and generate reliable future estimates.

A Linear Regression algorithm was employed for this purpose due to its effectiveness in modelling continuous numerical relationships and its interpretability in time-series forecasting scenarios. The model was trained separately for each currency to ensure accurate learning of currency-specific patterns and behaviours.

Feature Engineering

To enhance the predictive capability of the model, multiple time-dependent features were engineered from the historical data:

- Lag Features (lag1, lag2): Previous day and two-day prior exchange rates were used to capture short-term dependencies and sequential patterns.
- Moving Average (7-day): A rolling mean over a 7-day window was computed to smooth fluctuations and represent trend behaviour.

These features enabled the model to incorporate both immediate past values and short-term trends, improving its forecasting performance.

Model Training and Evaluation

The dataset was divided into training and testing subsets using an 80–20 split to evaluate the model's performance. The model was trained on the training set and validated on unseen test data.

To assess the effectiveness of the model, multiple evaluation metrics were used:

- Mean Absolute Error (MAE): Measures the average magnitude of prediction errors
- Root Mean Squared Error (RMSE): Penalizes larger errors more significantly
- Mean Absolute Percentage Error (MAPE): Provides error in percentage terms

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Future Forecasting

Using the trained model, future exchange rates were predicted for a period of 150 days. The prediction process was performed iteratively, where each predicted value was fed back into the model as input for subsequent predictions. This approach enabled the generation of continuous time-series forecasts for each currency.

The predicted values were then structured into a separate dataset and integrated with Power BI for visualization, allowing comparison between historical and forecasted trends.

Model Evaluation and Performance Metrics

To assess the effectiveness and reliability of the predictive model, several evaluation metrics were computed on the test dataset. These metrics provide a quantitative measure of the model's prediction accuracy and error levels.

The results obtained are as follows:

- Mean Absolute Error (MAE): 0.2716
- Root Mean Squared Error (RMSE): 0.3478
- Mean Absolute Percentage Error (MAPE): 0.2927%
- Model Accuracy: 99.70%

The low values of MAE and RMSE indicate that the model produces minimal prediction errors in actual units. Additionally, the very low MAPE value reflects high precision in percentage terms, resulting in an overall model accuracy of approximately 99.70%.

These results demonstrate that the model is highly effective in capturing exchange rate patterns and generating reliable forecasts.

Significance of the Model

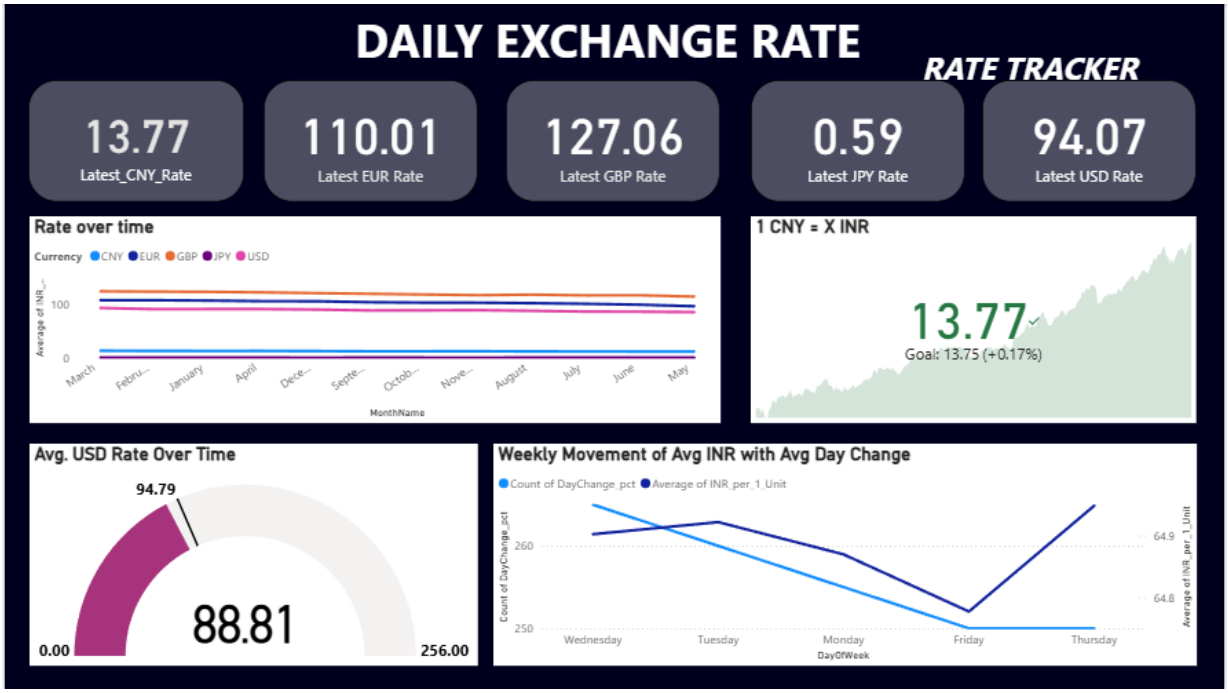
The implementation of the machine learning model adds a predictive dimension to the project, transforming it from descriptive analytics to advanced analytics. It enables:

- Anticipation of future currency trends
- Better understanding of market behaviour
- Support for data-driven decision-making

Overall, the model demonstrates how machine learning techniques can be effectively applied to financial time-series data to generate meaningful and actionable forecasts.

RESULTS AND KEY INSIGHTS

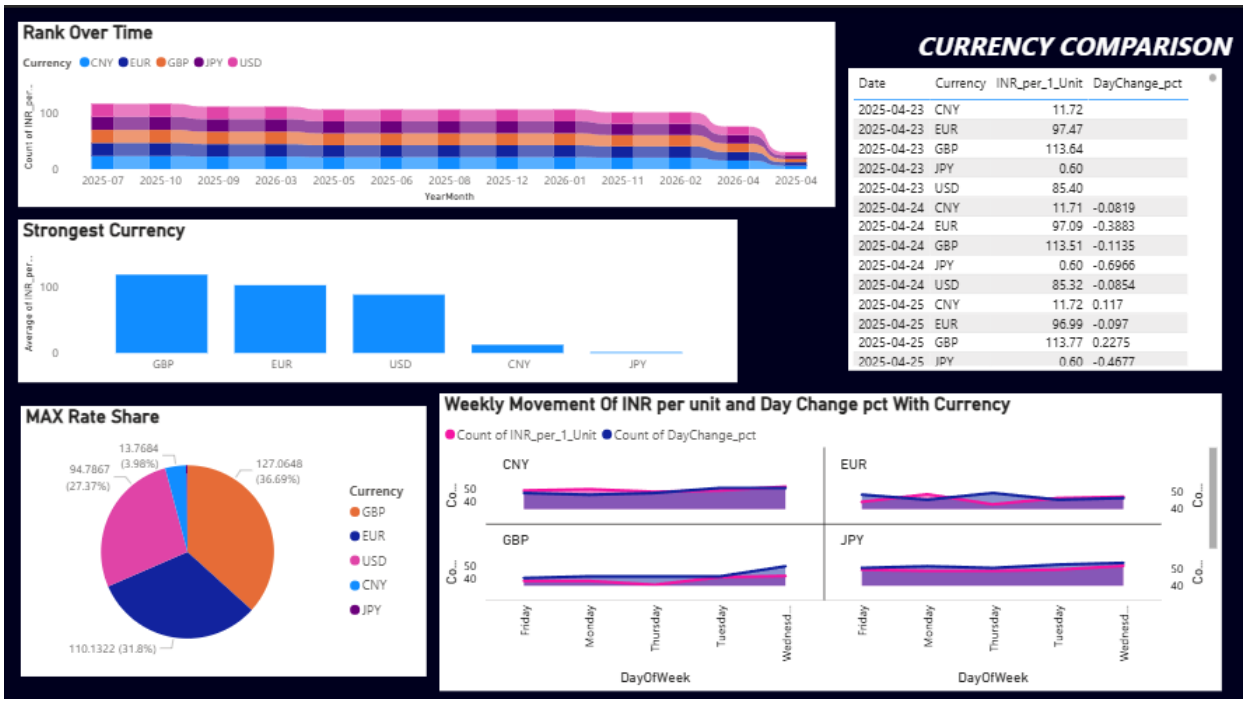
DASHBOARD1



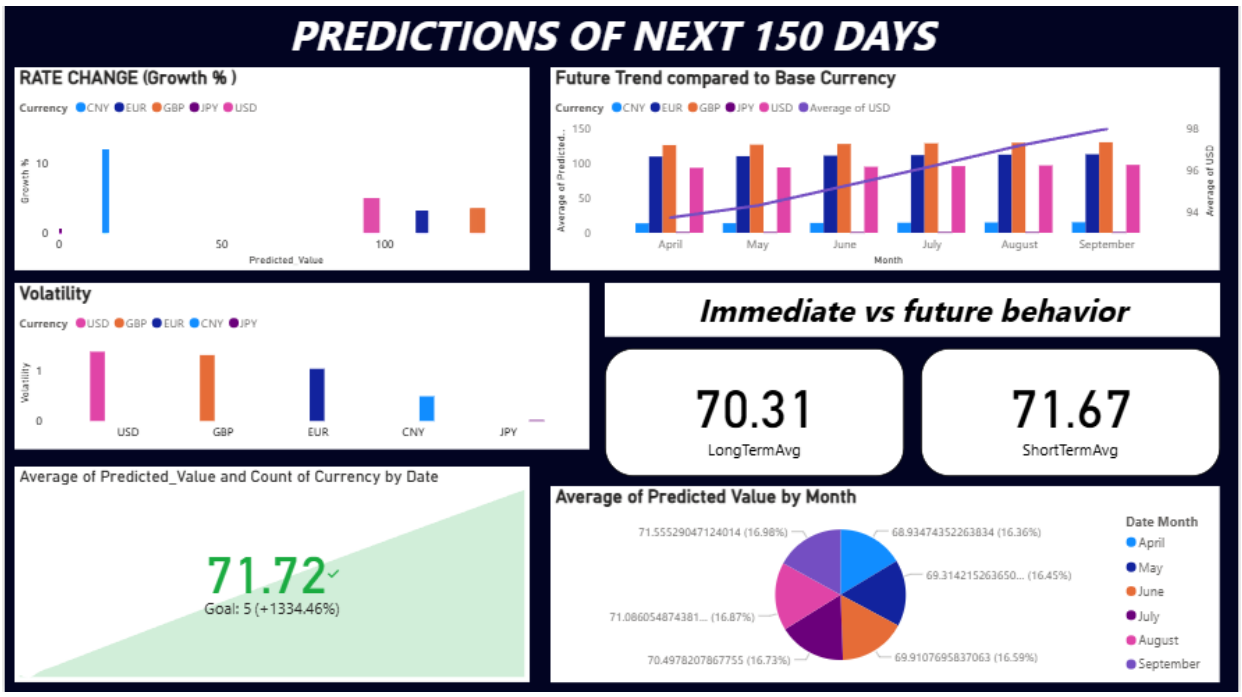
DASHBOARD2

DayOfWeek	CNY	EUR	GBP	JPY	USD	Total
Friday	1.097	1.1364	1.1723	1.6169	0.9857	**1.6169**
Monday	0.7203	-1.885	-1.3423	-2.5842	0.7339	**-2.5842**
Thursday	0.6964	1.3888	-1.23	-1.2019	0.8547	**1.3888**
Tuesday	-1.481	-1.9149	-1.4796	-2.0826	-1.5329	**-2.0826**
Wednesday	0.6827	-1.0593	-1.2062	1.3539	-0.8802	**1.3539**
Total	**-1.481**	**-1.9149**	**-1.4796**	**-2.5842**	**-1.5329**	**-2.5842**

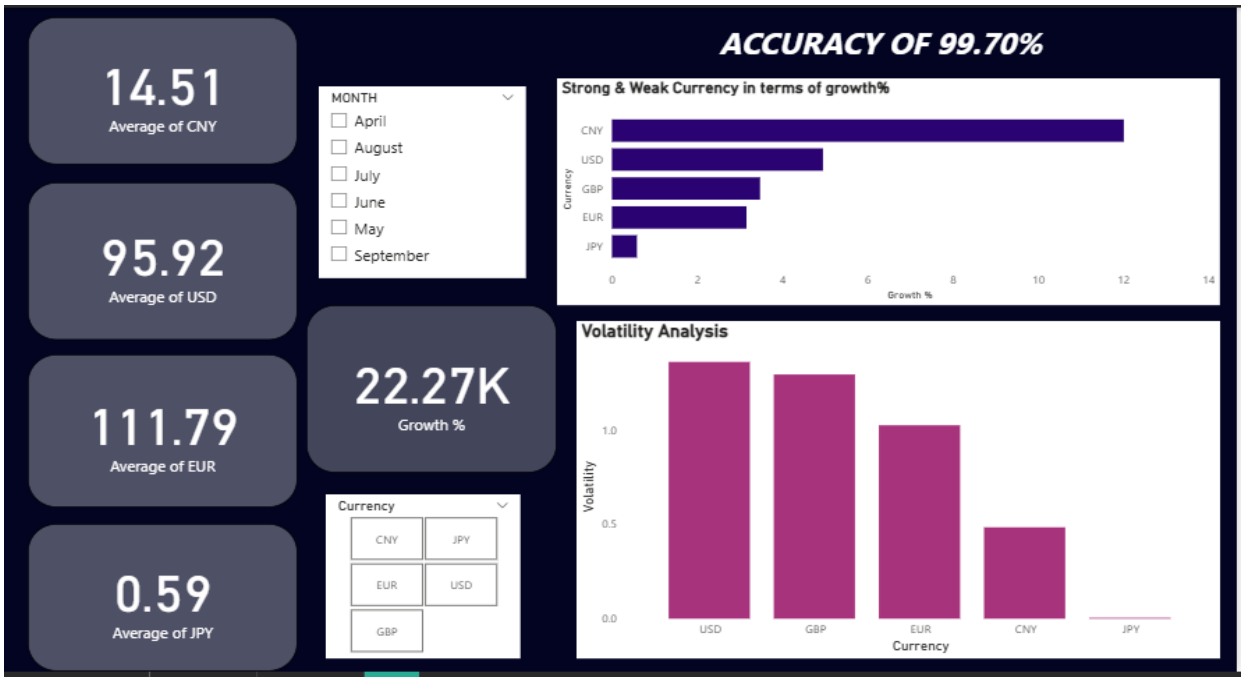
DASHBOARD3



DASHBOARD4



DASHBOARD5



KEY FINDINGS

- The analysis revealed clear trends in exchange rate movements of major currencies against INR over time.
- Comparative analysis showed differences in stability and fluctuation among currencies.
- Time-based analysis helped identify monthly and quarterly patterns in currency behaviour.
- Day-over-day percentage change highlighted short-term volatility in the market.
- The Power BI dashboards enabled interactive and effective visualization of currency trends.
- The machine learning model achieved high predictive performance (~99.70% accuracy), indicating reliable forecasting.
- The 150-day forecast provided a future outlook of exchange rate trends, supporting forward analysis.

APPLICATIONS

- The analysis can be used in financial decision-making, helping individuals and organizations understand currency trends for better planning.
- It supports import-export businesses in evaluating exchange rate movements and managing currency-related risks.
- This project can assist in investment analysis, where understanding currency behaviour is important for global investments.
- The predictive model enables forecasting of future exchange rates, which can be useful for budgeting and strategic planning.
- The dashboard provides a data-driven analytical tool for monitoring currency performance in an interactive and visual manner.

CONCLUSION

This project successfully demonstrates an end-to-end analytical approach to understanding and forecasting exchange rate movements. By transforming structured data into interactive Power BI dashboards, meaningful insights into currency trends and patterns were obtained.

The integration of a machine learning model further enhanced the project by introducing a predictive dimension, enabling the estimation of future exchange rates with high accuracy. The results indicate that the model is effective in capturing underlying patterns in the data and generating reliable forecasts.

Overall, the project highlights the effective use of data processing, visualization, and machine learning techniques to build a comprehensive analytical solution. It emphasizes the importance of data-driven approaches in analyzing financial trends and supports informed decision-making through both historical insights and future predictions.